



Iterative closest graph matching for non-rigid 3D/2D coronary arteries registration

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ABSTRACT

Background and objective Fusion of the preoperative computed tomography angiography and intraoperative X-ray angiography images can considerably enhance the visual perception of physicians during percutaneous coronary interventions. This technique can provide 3D information of the arteries and reduce the uncertainty of 2D guidance images. For this purpose, 3D/2D vascular registration with high accuracy and robustness is crucial for performing accurate surgery.

Methods In this study, we propose an iterative closest graph matching (ICGM) method that utilizes an alternative iteration framework including correspondence and transformation phases. A coarse-to-fine matching approach based on redundant graph matching is proposed for the correspondence phase. The transformation phase involves rigid and non-rigid transformations, in which rigid transformation is calculated using a closed-form solution, and non-rigid transformation is achieved using a statistical shape model established from a synthetic deformation dataset.

Results The proposed method is evaluated and compared with nine state-of-the-art methods on simulated data and clinical datasets. Experiments demonstrate that our method is insensitive to the pose of data and robust to noise and deformation. Moreover, it outperforms other methods in terms of registering real data.

Conclusions Given its high capture range, the proposed method can register 3D vessels without prior initialization in clinical practice.

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1. Introduction

Percutaneous coronary intervention is currently the main minimally invasive treatment for coronary artery disease, which is basically guided by X-ray angiography (XRA) images. For the lack of spatial information, interventional radiologists cannot easily recognize vessel branches, catheter tips, and narrowed coronary artery location in intraoperative XRA images. To address this issue, preoperative computed tomography angiography (CTA) imaging can be utilized in combination with intraoperative XRA imaging through 3D/2D registration technology. In this technique, the interventional image is augmented by fusing the projection of 3D vessels with 2D live images, which can provide depth information of vessels for the interventional radiologists and provide certain supports for instru-

ment operation under fluoroscopy images with low or no contrast agents.

3D/2D registration methods can generally be categorized into intensity-based, gradient-based, feature-based, and hybrid methods, according to the registration basis [1]. Intensity-based 3D/2D registration is achieved by comparing the similarity of projections of 3D pre-interventional [2] and 2D intra-interventional images and using an optimization method to refine the transformation parameters [3]. Gradient-based methods [4–6] utilize the gradients of 3D and 2D images to calculate similarities or correspondences. Feature-based methods [7–9] rely on matching features that can be extracted from CTA and XRA. Hybrid methods [10,11] combine intensity, gradient, or feature information to achieve 3D/2D registration.

Deep learning-based methods were explored recently in 3D/2D registration. Miao et al. [12] designed a nonlinear regressor to estimate transformation parameters using the residual between DRR

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and X-ray images. Convolutional neural networks (CNNs) are utilized to extract deep-level features from residual image patches and learn linear mapping from features to pose difference. Miao et al. [13] formulated 3D/2D registration as a Markov decision process with observations, actions, and rewards defined by the X-ray imaging systems. Multiple agents coupled with an auto attention mechanism and dilated fully convolution network are utilized for high robustness and efficiency. Liao et al. [14] proposed learning-based 3D/2D registration method in which the 2D point-to-point correspondences between pre- and intra-intervention images are established by a point-of-interest tracking network and then the 3D pose of pre-intervention data is estimated through a triangulation layer.

The most frequently discussed methods are feature-based methods due to the latter's ease of finding inherent correspondences between 3D and 2D features without prior projection. For instance, registration can be achieved by estimating the correspondence and transformation of fiducial markers implanted by an operator [15,16]. However, the marker-based methods are limited by the stability of implants, which heavily depend on the skill of the operator. Hence, intrinsic features in the image, such as anatomical points, geometrical structures, and image contours, are very useful for 3D/2D registration in clinical practice.

In 3D/2D registration of vessels, the centerlines are the most popular intrinsic features over a wide range. Baka et al. [7] introduced an extension of the iterative closest point (ICP) [17] for 3D/2D registration by back-projecting 2D points onto 3D space and then performing general ICP. Rivest-Hénault et al. [8] used a pre-computed distance transform of segmented vessels to construct an objective function, which accelerates the registration procedure. Similar to ICP, the iterative closest curve (ICC) [18] method utilizes vessel curves instead of points as elements to pair based on the closest relation and then computes the transformation by minimizing the distance between the paired curves. Given that the correspondence strategy is restricted to a one-to-one assignment, the ICP-like methods are sensitive to noise and outliers. The Gaussian mixture model (GMM) establishes a robust soft assignment for point set registration that relaxes the one-to-one correspondence to one-to-many in a probabilistic assignment framework [19–21]. Considering the nonlinear nature of perspective projections in 3D/2D point set registration, Kang et al. [9] extended the coherent point drifts method [19] by replacing the closed-form transformation solution in M-step of expectation maximization (EM) algorithm with particle swarm optimization (PSO). Baka et al. [7] proposed oriented GMM (OGMM) extended from [20] by incorporating orientational information of vessel points into 3D/2D coronary artery registration. These registration methods can achieve high registration accuracy given appropriate initial pose, while optimized similarity metrics produce a small capture range when aligning datasets with large-scale transformations.

Furthermore, the topology features of vessel centerlines provide effective information for calculating correspondences and registration. Benseghir et al. [22] proposed matching tree centerlines in a divide-and-conquer framework while ensuring the connectivity of bifurcation points. Liu et al. [23] regarded 3D and 2D vessels as tree topology and represented tree topology as a sequence; then, the vessel matching of nodes was achieved by performing sequence-to-sequence matching. The sequence was extracted using a topological sort algorithm by keeping the anteroposterior relationship and traversal in successive order.

In a registration task, rigid and non-rigid transformations are two successive procedures. The rigid registration can be separated as initial alignment and refined registration. Most above-mentioned methods [3,7,10,11,22,24] are focusing on accurate rigid registration given a tolerable initial pose. In general, pre-operative and intra-operative images are acquired from different equipments.

The capture range of most registration methods is insufficient to recover the transformation between the coordinate systems of two acquisition modalities. Markelj et al. [1] demonstrated that the 3D/2D registration can be initialized using several approaches, such as the alignment of patient position and orientation, the registration of corresponding pairs of markers, and manual initialization. However, an automatic initialization method that uses intrinsic features is more suitable for intraoperative 3D/2D registration. Varnavas et al. [25] precomputed 2D templates over a wide range of 3D poses and used an intra-operative pose to evaluate the similarity of the templates and generalized Hough transform of 2D fluoroscopy. Miao et al. [26] established a shape context encoded library of 2D contours extracted from the silhouettes of metal implants, and adopted Jensen-Shannon divergence as the matching metric for fast library matching.

Rigid transformation model is used in cerebral vasculature registration [3,11,24]. For coronary artery [7,8,23,27] and liver vessel registration [28,29], non-rigid deformation produced by respiratory and cardiac motion should be compensated to obtain high accuracy. Liu et al. [23] utilized 2D thin-plate spline to warp projected 3D vessel points to achieve non-rigid registration. However, non-rigid transformation on 3D space is more reasonable in 3D/2D registration. Rivest-Hénault et al. [8] formulated the energy for non-rigid transformation by composing displacement error, smooth constraint, and myocardium constraint in biplane fluoroscopy, thereby considerably reducing the ambiguity associated with monoplane scenario. Non-rigid 3D/2D registration of mono-plane angiogram remains an ill-posed problem. Hence, prior knowledge on the vessel structure must be employed to constrain deformations. For instance, Groher et al. [29] penalized changes in vessel length and unnatural bends in the energy function to generate admissible deformations of 3D vessel structures. Serradell et al. [30] took advantage of the recursive parameterization model of the coronary tree, which utilizes the dependencies between all nodes of a tree, to generate a synthetic deformation model. Baka et al. [31] proposed a method for building coronary motion models based on statistical shape models (SSM) generated from 3D+t CTA.

In this work, we propose an iterative closest graph matching (ICGM) method for 3D/2D coronary registration. This method utilizes an alternative iteration framework that includes correspondence and transformation phases. A coarse-to-fine matching is employed in the correspondence phase. Vessel node correspondences are obtained using the proposed redundant graph matching (RGM) method during coarse matching. Next, fine matching of vessel points is performed by assembling the edge-to-edge matching results based on node correspondences. Rigid and non-rigid transformations are carried out successively in the transformation phase. In rigid transformation, a fast closed-form solution [32] for the perspective- n -points (PnP) problem given n paired 3D and 2D points is used to obtain an approximate result. An optimization method is then used to refine the transformation. In non-rigid transformation, a 3D-to-2D model-fitting approach based on the synthetic deformation model of a coronary artery [30] is adopted.

The contributions of this work are threefold. First, an RGM method that can match two distinct graphs by constructing redundant cross-connected edges is proposed. Based on the RGM, a coarse-to-fine vessel matching method is also proposed to obtain accurate dense matching between 3D and 2D vessel points. Second, to reduce the dependency of the initial pose for rigid transformation, a fast closed-form solution with a high capture range is applied. To the best of our knowledge, this study is the first to apply the solution for the PnP problem to 3D/2D registration, because the proposed coarse-to-fine vessel matching addresses the problem of dense matching between 3D and 2D points. Third, an iterative framework is proposed to solve rigid and non-

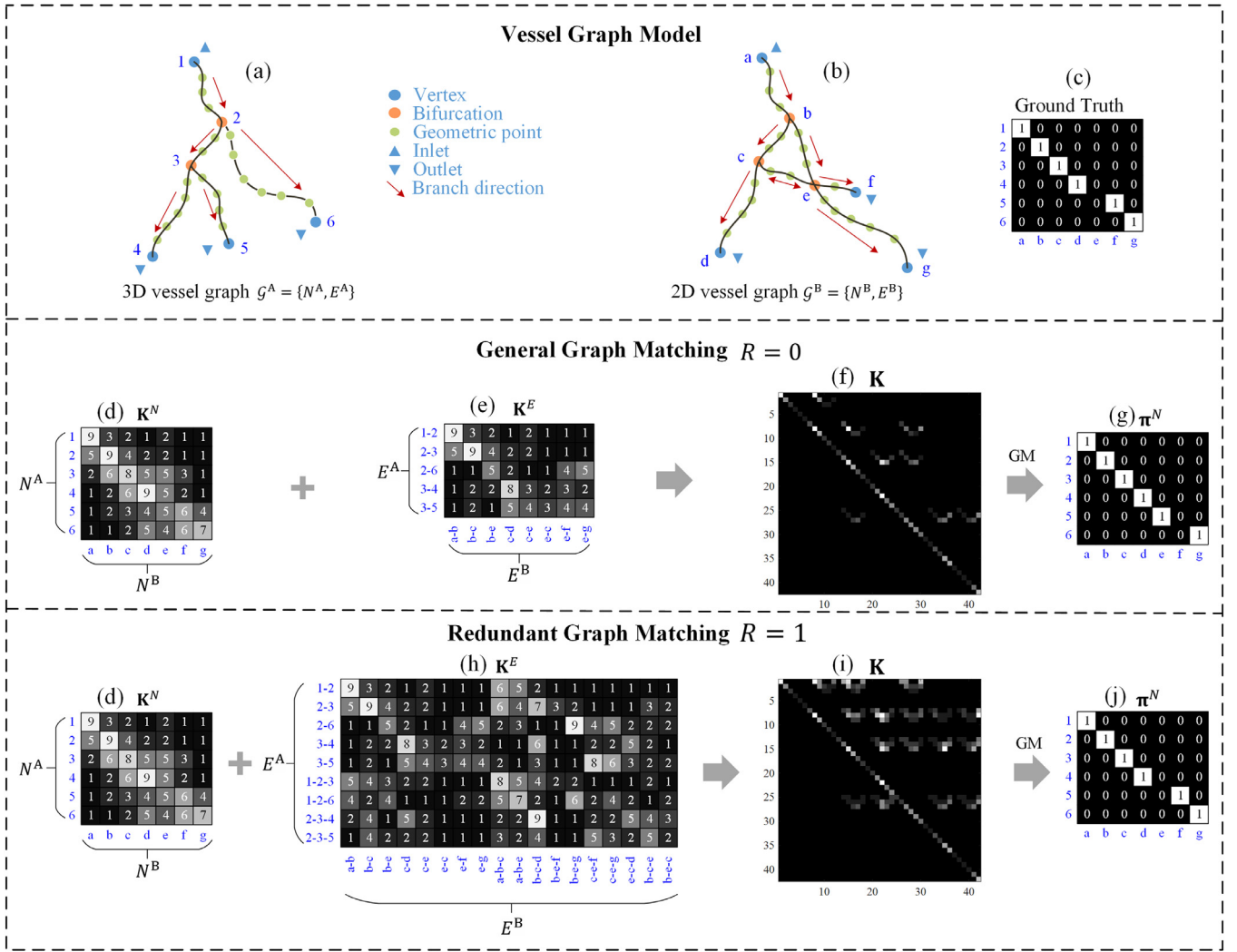


Fig. 1. Illustration of vessel graph and RGM. (a) 3D vessel graph $\mathcal{G}^A$ consisted of node set N^A and edge set E^A . (b) 2D vessel graph $\mathcal{G}^B$ consisted of node set N^B and edge set E^B . (c) The ground truth of node correspondence matrix between N^A and N^B . (d) The node affinity matrix. (e) The edge affinity matrix for general GM. (f) The affinity matrix for general GM. (g) The matching result of general GM. (h) The edge affinity for RGM ($R = 1$). (i) The affinity matrix for RGM ($R = 1$). (j) The matching result of RGM.

rigid vessel 3D/2D registration and compared with nine competing methods.

We evaluate the proposed method in simulated experiments under varying rotations, defects, noises, and deformations. As large pose differences are likely to occur during registration of multi-modality data, the purpose of the rotation tests is to evaluate the effective capture range of the proposed 3D/2D registration method. The noise and defect tests are performed to estimate the robustness of the method when the centerlines extracted from XRA images are imperfect due to stenosis, low contrast, and the presence of anatomical structures. Moreover, we evaluate the proposed method using real clinical datasets, including intra-patient CTA and XRA images; nine state-of-the-art 3D/2D registration methods are also compared with our method.

2. Material and methods

The vessel centerlines are represented in form of 3D or 2D graphs. A graph $\mathcal{G} = \{N, E\}$ in 3D and 2D is made up of nodes N and geometric edge set E connecting them. A graphical description of the vessel graph is provided in Fig. 1(a)-(b). 3D and 2D vessel graphs are denoted as $\mathcal{G}^A = \{N^A, E^A\}$ and $\mathcal{G}^B = \{N^B, E^B\}$ respectively. In these graphs, the node sets N^A and N^B contain vertices

and bifurcations, whereas the edge sets E^A and E^B are composed of geometric edges that connect two nodes. Furthermore, the vessel graph is modeled as a directed graph by specifying the vertex with the maximum radius as the inlet and the other vertexes as outlets. The direction of branches can be estimated by searching for directed flows from the inlet to the outlets. In 3D vessel graph, all the branches are unidirectional. By contrast, some edges are specified as bi-directional in 2D graph, as the edges $c-e$ and $e-c$ shown in Fig. 1(b).

2.1. Redundant graph matching

The RGM starts with the construction of redundant edge sets E^A and E^B , using the depth-first search to traverse the graph $\mathcal{G}^A$ and $\mathcal{G}^B$ by allowing a connection across R intermediate nodes. Vessel in 3D is considered an acyclic graph, whereas 2D vessel is generally represented by a cyclic 2D graph because several projected 3D branches will overlap in 2D, forming fake bifurcation and connecting edges. The cyclic 2D graph will hinder the subsequent matching procedure. A general approach for solving this problem is identifying fake bifurcations and disconnecting them to prevent cycle from forming the vessel graph [33]. However, overlapping on 2D

angiography is complicated, the disconnection of fake bifurcation may lead to inaccurate topology. In this study, we propose the RGM method to address the issue of matching two distinct graphs.

The general graph matching (GM) [34] is a quadratic assignment problem (QAP), which aims to find the optimal correspondence $\pi^N \in \{0, 1\}^{|N^A| \times |N^B|}$ between nodes by maximizing the following score [35],

$$\mathcal{S}(\pi^N) = \text{vec}(\pi^N)^T \mathbf{K} \text{vec}(\pi^N), \quad (1)$$

where π^N is a one-to-one mapping between node sets N^A and N^B , and $\text{vec}(\cdot)$ denotes the matrix vectorization. If a node $i \in N^A$ is matched with a node $j \in N^B$, then $\pi_{i,j}^N = 1$; otherwise, $\pi_{i,j}^N = 0$. The norm operator $|\cdot|$ denotes the number of elements in a set. The affinity matrix $\mathbf{K} \in \mathbb{R}^{|N^A| \times |N^B| \times |N^A| \times |N^B|}$ can be formulated from two affinity matrices $\mathbf{K}^N \in \mathbb{R}^{|N^A| \times |N^B|}$ and $\mathbf{K}^E \in \mathbb{R}^{|E^A| \times |E^B|}$ [34], which are specified to estimate the similarity of nodes and edges, respectively.

Illustration of general GM and RGM is shown in Fig. 1. In general GM, the edges are constructed using branches that directly connect two nodes. Edge 3-5 of E^A should match edge $c-f$ of E^B . However, fake bifurcation e of N^B may hamper matching. Therefore, we construct edges by allowing a cross-connection between two nodes, also called as a redundant edge. In this manner, edge 3-5 can match redundant edge $c-e-f$ when the affinity value is set. A redundant degree R , which denotes the number of intermediate nodes that can be crossed to link two nodes, is applied to control the complexity of RGM. The affinity matrix $\mathbf{K}$ (Fig. 1(f)) is constructed from $\mathbf{K}^N$ (Fig. 1(d)) and $\mathbf{K}^E$ (Fig. 1(e)). In the illustration, we manually set the affinity value of nodes and edges as integer from 1 to 9. The pairing result of general GM is obtained by solving the QAP (1), as shown in Fig. 1(g). Results reveal that node 5 of N^A is mistakenly matched with node e of N^B . In RGM, abundant edges are constructed, as shown in Fig. 1(h). The affinity matrix $\mathbf{K}$ (Fig. 1(i)) of RGM is denser than that of general GM. The pairing result of RGM is shown in Fig. 1(j). Compared with the ground truth (Fig. 1(c)) of the node correspondence between N^A and N^B , the RGM ($R = 1$) can achieve a better matching result than general GM ($R = 0$).

The difference between RGM and general GM is the construction of the edge set, the redundant edge set of RGM carries out a denser affinity matrix compared with that obtained through general GM. Denser affinity matrix enables more possibilities for correct matching. Theoretically, a higher R will produce more accurate results. However, it also increases computational burden and that forms a trade-off. The illustration also reveals that the RGM can produce accurate result when given the accurate affinity matrices $\mathbf{K}^N$ and $\mathbf{K}^E$. Therefore, the remaining question of RGM will focus on the construction of node and edge affinity matrices. In our definition, the redundant edge set contains directed connected edges. In the following sections, all edges are regarded as redundant edge.

2.2. Coarse-to-fine vessel matching

A coarse-to-fine vessel matching is introduced in this section. In coarse matching, the node correspondences π^N can be obtained using RGM. Then the fine matching $\pi \in \{0, 1\}^{|P^A| \times |P^B|}$ of 3D points $P^A = \{\mathbf{p}_i^A\}$ and 2D points $P^B = \{\mathbf{p}_j^B\}$ can be obtained by assembling the edge-to-edge matching results based on the node correspondences.

Then the remaining problems will focus on the setting of the node affinity matrix $\mathbf{K}^N$ and edge affinity matrix $\mathbf{K}^E$. Two attributes of graph node, point coordinates $\mathbf{p}$ and connectivity degree ρ , are considered to calculate the node affinity. The distance can be calculated between the projected point $\text{Proj}(\mathbf{p}_i^A)$ and 2D point $\mathbf{p}_j^B$, where $\text{Proj}(\cdot)$ denotes perspective projection operator, i and j de-

note the indexes of nodes. The projection operator is expressed as a projection matrix which can be build using several parameters obtained from the DICOM head file of XRA. The connectivity degree ρ is employed to distinguish vertex ($\rho = 1$) and bifurcation ($\rho \geq 3$), which is defined as how many nodes are connected with this node directly. The $\mathbf{K}^N$ can be formulated as

$$\mathbf{K}^N = \exp\left(-\frac{\mathbf{D}^p}{\sigma_p}\right) + \exp\left(-\frac{\mathbf{D}^\rho}{\sigma_\rho}\right), \quad (2)$$

where the discrepancy matrices $\mathbf{D}^p, \mathbf{D}^\rho \in \mathbb{R}^{|N^A| \times |N^B|}$ encode the pairwise L_2 norm between two sets of node coordinates and connectivity degrees respectively. The edge affinity is specified using two items, edge length l and the sequential point set of edge s . The $\mathbf{K}^E$ is formulated as follows:

$$\mathbf{K}^E = \exp\left(-\frac{\mathbf{D}^l}{\sigma_l}\right) + \exp\left(-\frac{\mathbf{D}^s}{\sigma_s}\right). \quad (3)$$

The edge of 3D graph is projected onto 2D plane. Then, the geodesic distances l_u^A of the projected 3D edge $\text{Proj}(e_u^A)$ and l_v^B of 2D edge e_v^B are calculated, where u and v denote the indexes of edges. $\mathbf{D}^l \in \mathbb{R}^{|E^A| \times |E^B|}$ denotes the pairwise discrepancy between two sets of edge length using absolute difference. $\mathbf{D}^s \in \mathbb{R}^{|E^A| \times |E^B|}$ denotes the pairwise discrepancy of two sequential point sets $\text{Proj}(s_u^A)$ and s_v^B using Fréchet distance [36]. The normalization factors $\sigma_p, \sigma_\rho, \sigma_l, \sigma_s$ are the standard deviations estimated from the off-diagonal elements of associated discrepancy matrices.

Dense correspondences are required for accurate 3D/2D vessel registration. Given the result of node pairing π^N , the fine matching π can be achieved by assembling correspondences between branch points that lie on paths linking matched nodes. In practice, we consider the edge of a 3D vessel as a reference to find an edge that connects two matched nodes in a 2D vessel. For an arbitrary edge $e_u^A \in E^A$, if an edge $e_v^B \in E^B$ matches it, then only condition $\pi_{u^1, v^1}^N \cdot \pi_{u^\#, v^\#}^N = 1$ is satisfied, where u^1, v^1 denote the starting point of the edge and $u^\#, v^\#$ denote its ending point. If a loop exists in the 2D graph, multiple candidate edges are produced to match with a 3D edge, as shown in Fig. 1. We utilize discrete Fréchet distance to determine the most likely path to improve the robustness of the proposed method. The correspondences between a projected 3D edge and its paired 2D edge (or redundant edge) can be achieved by dynamic time wrap (DTW) [37].

The details of coarse-to-fine vessel matching can be summarized in Table 1.

2.3. 3D/2D registration

Fusing the 3D vessel structure $\mathcal{G}^A$ and the 2D vessel structure $\mathcal{G}^B$ can be generally formulated as finding the optimal transformation T^* of the 3D vessel that best aligns the 2D vessel. For the graph $\mathcal{G}^A$, the operation of transformation and projection will not change the topology of the points. In general, an intuitive estimation of the registration result is based on the sum of Euclidian distances of the corresponding point pairs. The registration procedure can thus be reformulated as follows

$$\{T^*, \pi^*\} = \arg \min_{T, \pi} \left\{ \sum_i \sum_j \pi_{ij} d(\text{Proj}(T(\mathbf{p}_i^A)), \mathbf{p}_j^B) \right\}, \quad (4)$$

where $\mathbf{p}_i^A \in \mathbb{R}^3 \times 1$ and $\mathbf{p}_j^B \in \mathbb{R}^2 \times 1$ are the points on 3D and 2D vessel centerline, $d(\cdot)$ denotes Euclidian distance, $\text{Proj}(\cdot)$ denote perspective projection.

To solve (4), we apply the ICP-like iterative framework by alternately establishing the correspondences π and estimating the transformation. For a given transformation, optimization of (4) can

Table I

Pseudocode of the coarse-to-fine vessel matching algorithm.

Input: vessel graph of 3D $\mathcal{G}^A$ and 2D $\mathcal{G}^B$, projection operator *Proj*.
Parameters: the degree R for RGM.
Output: the final pairing matrix π .
1: Construct redundant edge sets of $\mathcal{G}^A$ and $\mathcal{G}^B$ with degree R ;
2: Calculate node affinity matrix $\mathbf{K}^N$ in (2) and edge affinity $\mathbf{K}^E$ in (3);
3: Solve (1) to get coarse matching matrix π^N ;
4: Find paired 2D edge for each 3D edge;
5: Get fine matching π by assembling each edge curve matching result calculated using DTW;

Table II

Pseudocode of 3D/2D vessel structure registration.

Input: vessel graph of 3D $\mathcal{G}^A$ and 2D $\mathcal{G}^B$, projection operator *Proj*.
Parameters: the degree R for RGM, parameters to control synthetic deformation prior of coronary $\sigma_r, \sigma_\theta, \sigma_\psi$, parameter to control SSM deformation scale β .
Output: registration result, namely the 3D transformation T and deformation parameter $\mathbf{b}$.
Rigid Registration:
1: Repeat
2: Perform the coarse-to-fine graph matching in Table I;
3: Calculate 3D transformation T using a fast closed-form solution [33];
4: Refine the 3D transformation based on a quasi-newton algorithm solving (5);
5: Until Converge
Non-rigid Registration:
6: Establish a synthetic SSM from the rigid registered 3D vessel centerline;
7: Repeat
8: Perform the coarse-to-fine graph matching in Table I;
9: Calculate deformation parameter $\mathbf{b}$ using SSM fitting in (7);
10: Until Converge

be solved by the proposed coarse-to-fine vessel matching, introduced in section B. Given the correspondence, the remaining problem focuses on finding a rigid or non-rigid transformation T^* that best aligns corresponding the 3D and 2D point set,

$$T^* = \arg \min_T \sum_i d(\text{Proj}(T(\mathbf{p}_i^A)), \mathbf{p}_{\pi(i)}^B), \quad (5)$$

the pairing matrix π is rewrote as a pairing mapping $\pi(i) = j$ if $\pi_{ij} = 1$.

2.3.1. Rigid transformation

For rigid transformation T_{rigid} , large capture range and high accuracy are required. In 3D/2D registration, there often exists large pose difference between the multi-modality data. The transformation T_{rigid} includes 3 translations and 3 rotations. A fast closed-form solution with $O(n)$ time complexity is employed here [32], where n denotes the counts of points. This method formulated the coordinates of n 3D points $\mathbf{p}_i^A$ as weighted sums of four virtual control points $\{\mathbf{v}_j\}$, $\mathbf{p}_i^A = \sum_{j=1}^4 \alpha_{ij} \mathbf{v}_j$, with $\sum_{j=1}^4 \alpha_{ij} = 1$. Supposing each projection of 3D point coincides with the corresponding 2D point $\text{Proj}(\sum_{j=1}^4 \alpha_{ij} T_{\text{rigid}}(\mathbf{v}_j)) = \mathbf{p}_{\pi(i)}^B$. Then we can derive

$$\omega_i \begin{bmatrix} \mathbf{p}_{\pi(i)}^B \\ 1 \end{bmatrix} = \mathbf{H} T_{\text{rigid}}(\mathbf{p}_i^A) = \mathbf{H} \sum_{j=1}^4 \alpha_{ij} T_{\text{rigid}}(\mathbf{v}_j), \quad (6)$$

where $\mathbf{H} \in \mathbb{R}^{3 \times 3}$ is the internal calibration matrix of the camera and ω_i is homogeneous coefficient. The solution of $\{T_{\text{rigid}}(\mathbf{v}_j)\}$ can be achieved by solving a small constant number of linear equations with $O(n)$ time complexity.

This method transfers the problem of finding a best rigid transformation to estimating the coordinates of four control points $\{T_{\text{rigid}}(\mathbf{v}_j)\}$, and the rigid transformation T_{rigid} can be obtained using a least-squares method. The closed-form solution has a large capture range, but it loses accuracy. Therefore, a refinement procedure is performed after the closed-form solution to increase the accuracy. We use a quasi-newton algorithm to achieve refinement of the 3D transformation T_{rigid} by optimizing (5).

2.3.2. Non-rigid transformation

We propose a 3D-to-2D model fitting approach based the synthetic deformation prior of coronary tree [30] to obtain non-rigid transformation T_{nonrigid} . Given a 3D coronary tree $\mathcal{G}^A$ and its centerline points $\mathbf{P}^A \in \mathbb{R}^{|\mathbf{P}^A| \times 3}$, we can generate a population of synthetic deformed vessel centerlines with the same anatomical correspondence as $\mathbf{P}^A$. The deformation model can be obtained by establishing the following SSM $T_{\text{nonrigid}}(\mathbf{P}^A; \mathbf{b}) = \tilde{\mathbf{P}} + \Phi \mathbf{b}$, where $\mathbf{P}^A \in \mathbb{R}^{3|\mathbf{P}^A| \times 1}$ denotes the vectorization of matrix $\mathbf{P}^A$, the vectorized average shape $\tilde{\mathbf{P}}$ and the main modes that make up matrix Φ are precomputed from the generated training data set and $\mathbf{b}$ is a vector that parameterizes the non-rigid transformation T_{nonrigid} . We force the parameters to stay within a range to generate plausible variations. For instance, each parameter b_i in vector $\mathbf{b}$ is forced to lie within $-\beta\sqrt{\lambda_i} \leq b_i \leq \beta\sqrt{\lambda_i}$, where λ_i is the i th element of λ , and the eigenvalue vector of the covariance matrix of the training shapes and β controls the range of hype-rectangle.

Each 2D point $\mathbf{p}_{\pi(i)}^B$ is back-projected onto 3D space along a casting ray defined by the X-ray source and the 2D point in imaging plane. Given its paired 3D point $\mathbf{p}_i^A$, we can determine a 3D point $B\text{Proj}(\mathbf{p}_{\pi(i)}^B)$ closest to the casting ray, where $B\text{Proj}(\bullet)$ denotes the back-projection operation. We rewrite the correspondence permutation of the 2D point set as $(\mathbf{p}^B) \in \mathbb{R}^{2 \times |\mathbf{P}^A|}$. If $T_{\text{nonrigid}}(\mathbf{P}^A; \mathbf{b}) = B\text{Proj}(\pi(\mathbf{p}^B))$ is satisfied, then, $\mathbf{b}$ must be optimal. Therefore, we can fit the established SSM to the back-projected points to obtain the optimal $\mathbf{b}^*$. By using the least-squares approximation method to calculate the shape parameter adjustments and restricting $\mathbf{b}$ to a hyper-rectangle, we derive:

$$\begin{aligned} \mathbf{b}^* &= \mathbf{b} + d\mathbf{b} \\ d\mathbf{b} &= \Phi^T (\text{vec}(B\text{Proj}(\pi(\mathbf{p}^B))) - \text{vec}(\mathbf{P}^A)), \\ \forall i, -\beta\sqrt{\lambda_i} &\leq b_i \leq \beta\sqrt{\lambda_i} \end{aligned} \quad (7)$$

where $\text{vec}(\bullet)$ denotes vectorization of matrix, $d\mathbf{b}$ denotes differential of vector $\mathbf{b}$.

The algorithm for 3D/2D vessel structure registration can be summarized as the following pseudocode (Table II).

Table III
Parameters used in all 3D/2D registration methods.

	stop criterion		Optimizer	SSM param calculation
	max iterations	counts		
ICP-BP	300	10^{-5}	Levenberg-Marquardt	3D-to-2D fitting
ICP-PnP	300	10^{-5}	PnP	3D-to-2D fitting
DT	300	10^{-5}	Levenberg-Marquardt	optimization
ICC	10	10^{-5}	Quasi-Newton	3D-to-2D fitting
Tree	10	10^{-5}	Quasi-Newton	3D-to-2D fitting
GMM	100	10^{-5}	Quasi-Newton	optimization
OGMM	100	10^{-5}	Quasi-Newton	optimization
PSO-EM	500	10^{-5}	PSO	optimization
Grad	100	10^{-5}	Quasi-Newton	optimization
ICGM	10	10^{-5}	PnP & Quasi-Newton	3D-to-2Dfitting

2.4. Dataset and preprocess

The dataset consists of intra-patient CTA and XRA coronary images, which were collected in Peking Union Medical College Hospital. Totally 60 pairs of 3D and 2D vessel data, involving 32 pairs of right coronary arteries (RCA) and 28 pairs of left coronary arteries (LCA), are used in this paper.

The CTA images and XRA sequences are preprocessed to obtain the 3D graph G^A and 2D graph G^B . For each CTA image, the medical image processing software (*Materilise Mimics*) is used to obtain the centerlines and the connection information manipulated by doctors. The XRA images are enhanced using a background subtraction method [38] and segmented by a threshold. Then, centerlines are extracted by applying iterative filtering operation and multi-direction indexed non-maximum suppression [39,40]. An eight-neighborhood tracking method is then used to connect adjacent 2D points. Vessel segmentation, centerline extraction, and graph construction can be replaced by other methods. In the experiment, all comparative methods are performed on the same vessel graphs.

The counts of 3D vessel graph node N^A range from 16 to 23 for LCA and from 8 to 15 for RCA. The number of sampled 3D vessel points P^A is approximately 500. The counts of 2D vessel graph node N^B range from 22 to 35 for LCA and from 16 to 23 for RCA. The number of sampled 2D vessel points P^B is approximately 1000. Directed graphs are constructed by specifying the inlet, which can be automatically recognized by selecting the vertex with the maximum radius.

We design simulated and real clinical data experiments to evaluate the performance of the proposed 3D/2D registration method compared with several state-of-the-art methods. The 3D/2D registrations are evaluated by calculating the mean projected distance (mPD) [41] between projected 3D and 2D centerline points. For the real-data experiments, we manually record the correspondences of node points between 3D and 2D vessel graphs to compute mPD. For the simulation experiments, the correspondences of node points between 3D and 2D graphs are automatically recorded during the data generation procedure.

The performance of the proposed 3D/2D registration method is compared with nine competing methods in each set of experiments.

1) 3D/2D ICP-BackProjection (ICP-BP for short), which uses a back-projecting strategy, extends ICP to the 3D/2D application introduced in [7]. 2) An extension of ICP using the closed-form solution of PnP to calculate the transformation matrix (ICP-PnP for short) is first proposed in this paper. ICP-PnP is performed in an iterative procedure: assigning a 2D centerline point to each projected 3D centerline point based on the closest distance relation, solving the transformation using the PnP strategy, updating the projections of 3D points using rigid transformation. 3) An accelerated method for ICP matching (DT for short) [8], using a pre-computed distance transformation of a segmented vessel tree. 4)

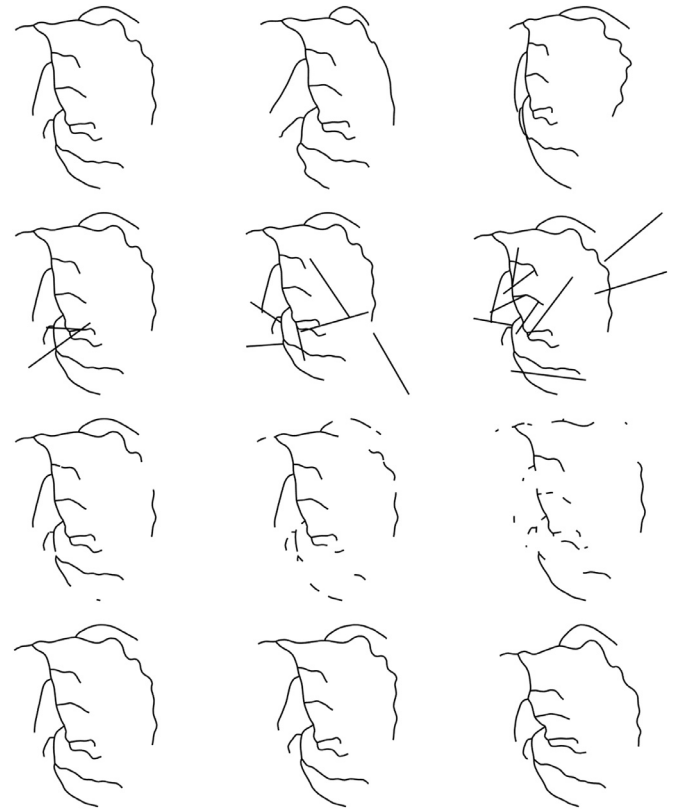


Fig. 2. Examples of the simulated data. First row: Complete projected vessel centerline, transformed centerline with $\sigma_\theta = 10^\circ$, transformed centerline with $\sigma_\theta = 30^\circ$. Second row: centerline with $p_n = 10\%$ adding noise, centerline with $p_n = 30\%$ adding noise, centerline with $p_n = 50\%$ adding noise. Third row: centerline with $p_d = 10\%$ missing points, centerline with $p_d = 30\%$ missing points, centerline with $p_d = 50\%$ missing points. Fourth row: deformed vessel centerline $\sigma_b = 0.1$, deformed vessel centerline $\sigma_b = 0.5$, deformed vessel centerline $\sigma_b = 1.0$.

The curvilinear structure pairing framework (ICC for short) is applied to 3D/2D coronary artery registration [18]. 5) Tree topology matching for 3D/2D registration (Tree for short) involves a class of methods using tree topology consistency matching [22]. 6) 3D/2D GMM [20] and 7) OGMM [7]. The Gaussian scales of pixels in GMM and OGMM are 1, and the scale of orientation in OGMM is 0.2. 8) PSO-EM [9], which is designed for rigidly registering 3D and 2D feature points in an expectation-maximization framework. 9) A hybrid-based 3D/2D rigid registration method (Grad for short) of cerebral vessel using centerline of 3D vessel and gradient of 2D image [11]. For it is based on the gradient of 2D image, we only test it on real data.

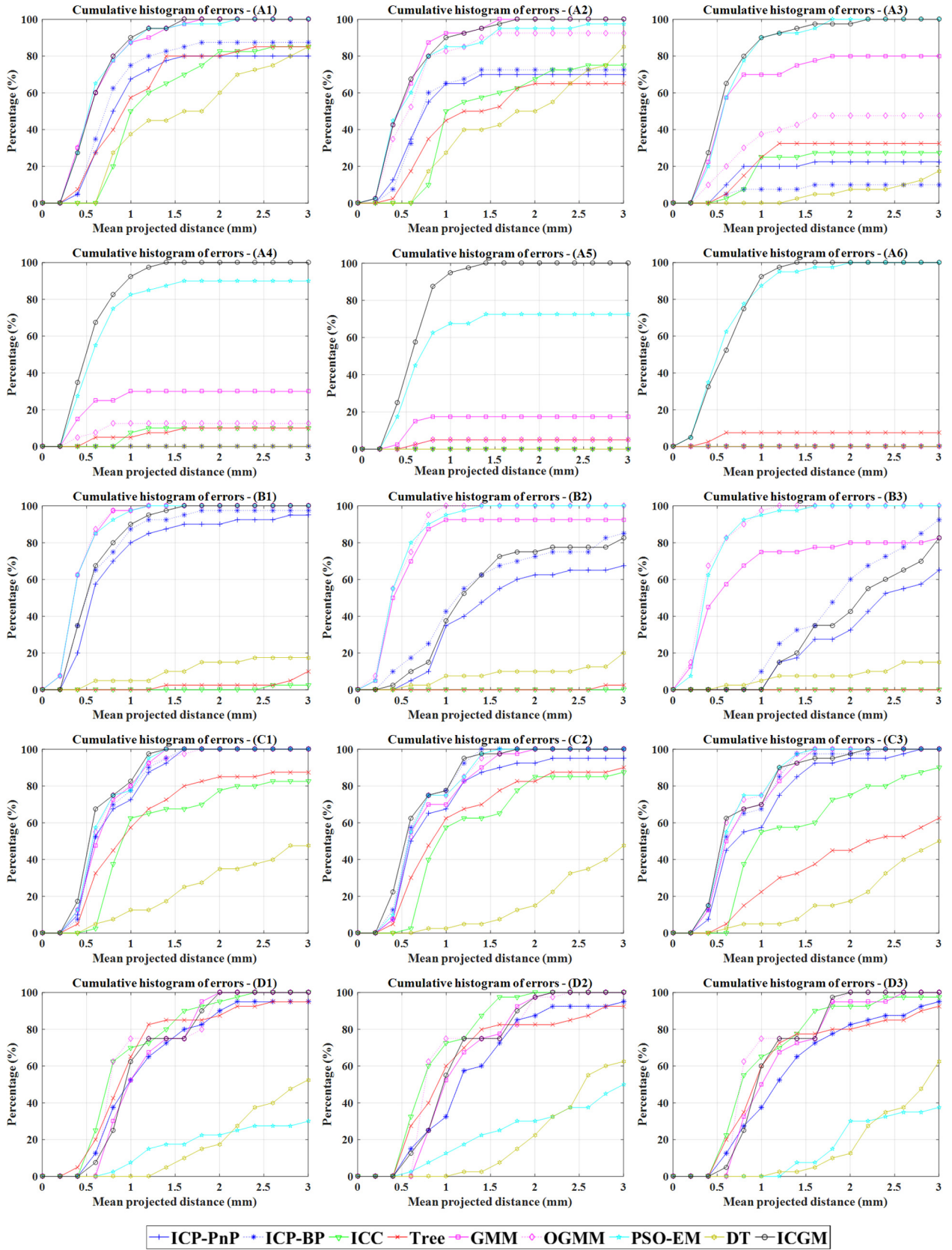


Fig. 3. Results of simulated rigid and non-rigid registration experiments with varying rotation angle differences, segment defects, branch noises and deformation. 8 kinds of comparing methods and the ICGM are tested. Percentage is indicated by Y-axis and 100% percentage indicates 80 times registrations. Mean projected distance (unit: mm) is indicated by X-axis. The parameters of simulated data are: (A1) $\sigma_\theta = 10^\circ$, (A2) $\sigma_\theta = 30^\circ$, (A3) $\sigma_\theta = 60^\circ$, (A4) $\sigma_\theta = 90^\circ$, (A5) $\sigma_\theta = 120^\circ$, (A6) $\sigma_\theta = 180^\circ$, (B1) $p_d = 10\%$, (B2) $p_d = 30\%$, (B3) $p_d = 50\%$, (C1) $p_n = 10\%$, (C2) $p_n = 30\%$, (C3) $p_n = 50\%$, (D1) $\sigma_b = 0.1$, (D2) $\sigma_b = 0.5$, (D3) $\sigma_b = 1.0$.

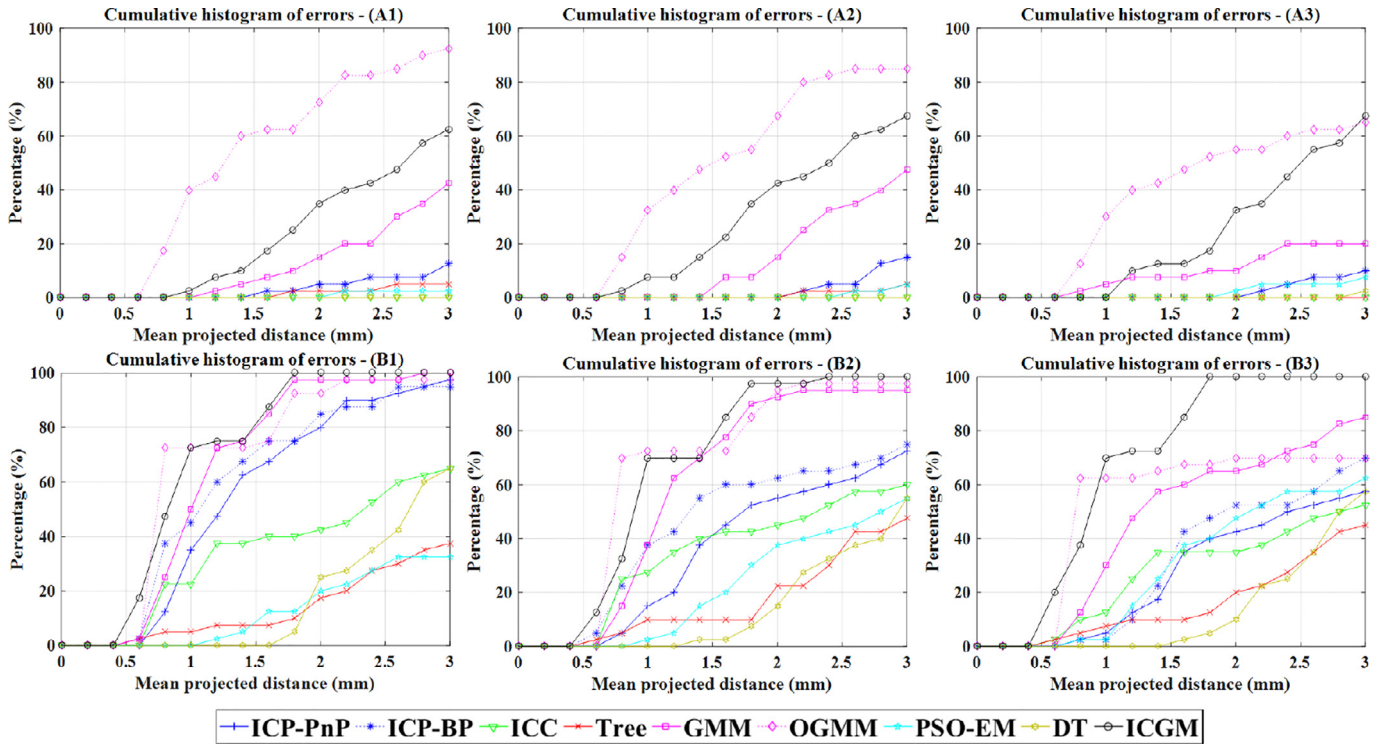


Fig. 4. Results of simulated non-rigid registration experiments with combined defects. 8 kinds of comparing methods and the ICGM are tested. Percentage is indicated by Y-axis and 100% percentage indicates 80 times registrations. Mean projected distance (unit: mm) is indicated by X-axis. The parameters of simulated data are: (A1) $\sigma_\theta = 10^\circ$, $\sigma_b = 1.0$, $p_d = 30\%$; (A2) $\sigma_\theta = 30^\circ$, $\sigma_b = 1.0$, $p_d = 30\%$; (A3) $\sigma_\theta = 60^\circ$, $\sigma_b = 1.0$, $p_d = 30\%$; (B1) $\sigma_\theta = 10^\circ$, $\sigma_b = 1.0$, $p_n = 30\%$; (B2) $\sigma_\theta = 30^\circ$, $\sigma_b = 1.0$, $p_n = 30\%$; (B3) $\sigma_\theta = 60^\circ$, $\sigma_b = 1.0$, $p_n = 30\%$.

These methods are performed until a stopping criterion, such as a maximum number of iterations or parameters change less than a threshold, is satisfied, as shown in Table III.

The proposed iterative closest graph matching for 3D/2D registration is called ICGM for short. The redundant degree $R = 2$ is applied empirically. For all comparing methods, non-rigid transformation is parameterized by an SSM constructed using a synthetic deformation model. The parameters used to generate the synthetic datasets involves three standard deviations $\sigma_r = 1, \sigma_\theta = \sigma_\varphi = 0.5$ [30], 5000 deformed samples are generated for each 3D vessel. For matching-based methods (ICGM, ICP-PnP, ICP-BP, ICC, and Tree), the deformation parameter $\mathbf{b}$ is calculated using the 3D-to-2D SSM fitting in (7). For optimization-based methods (GMM, OGMM, PSO-EM, DT and Grad), the deformation parameter $\mathbf{b}$ is updated to optimize their cost functions. For all methods, $\beta = 3$ is applied to restrict vessel deformations to a plausible range.

2.5. Experimental design

The simulated data are generated from 3D centerlines by considering four aspects: rigid transformation, non-rigid deformation, branch noise, and segment defect. For rigid transformation, most existing methods are capable of aligning data with small rotation angles but are incapable of large-scale transformation. The proposed registration method based on RGM is designed for registering data with an arbitrary rotation angle. The 3D vessel structure $\mathcal{G}^A$ is projected onto a 2D plane defined by the camera internal matrix from a XRA dicom info file to obtain the simulated 2D vessel structure $\mathcal{G}^B = \text{Proj}(\mathcal{G}^A)$.

In the experiment, four 3D vessels are used to generate simulated 2D vessels. For each set of generation parameters, 20 2D vessels are generated; that is, 80 registrations are performed for each method.

To evaluate the ability of aligning rigid transformation, a specific rigid transformation is applied on 3D vessel structure to get $T(\mathcal{G}^A, \theta)$. In this work, only rotation is considered, as translation error can be eliminated using a translation adjustment operation. We randomly set the rotation angle θ following a normal distribution $\theta \sim \mathcal{N}(\sigma_\theta, 5^\circ)$ and one random axis is selected to rotate the points. The parameters $\{\sigma_\theta\}$ are set as $\{10^\circ, 30^\circ, 60^\circ, 90^\circ, 120^\circ, 180^\circ\}$.

For non-rigid transformation, the synthetic deformation model is employed to warp the 3D vessel structure. The SSM is established from synthetic data to retain 95% of the variance, resulting in approximately 20 parameters. In the phantom experiment, the deformation model parameter is randomly assigned following the normal distribution $\mathbf{b} \sim \mathcal{N}(\sigma_b, 0.05) \times 3\sqrt{\lambda}$, where λ is the eigen values. The parameters are $\{\sigma_b\} = \{0.1, 0.5, 1.0\}$.

Branch noise and segment defects are randomly added to the 2D centerlines to simulate the real centerline of vessels. We use a percentage value to specify the parameter of branch noise and segment defect, which means a certain percentage p_n or p_d of points are added to or deleted from the 2D vessel structure. When adding the branch noise, randomly generated free form implies that the topology of the 2D vessel is changed. When deleting a segment, a length of points of arbitrary branch are selected, to simulate the vascular stenosis in patient XRA. In the phantom experiment, we set $\{p_n\} = \{10\%, 30\%, 50\%\}$ and $\{p_d\} = \{10\%, 30\%, 50\%\}$ to perform the registration. Example of simulations of rigid transformation, non-rigid deformation, branch noise and segment defect are demonstrated in Fig. 2.

In the real-data experiments, the initial projection of the 3D vessel onto the 2D image plane is consistently located far from the 2D vessel and exhibits large rotation angle discrepancies. For pose initialization depended methods, although the translation between 3D and 2D data can be adjusted by a normalization opera-

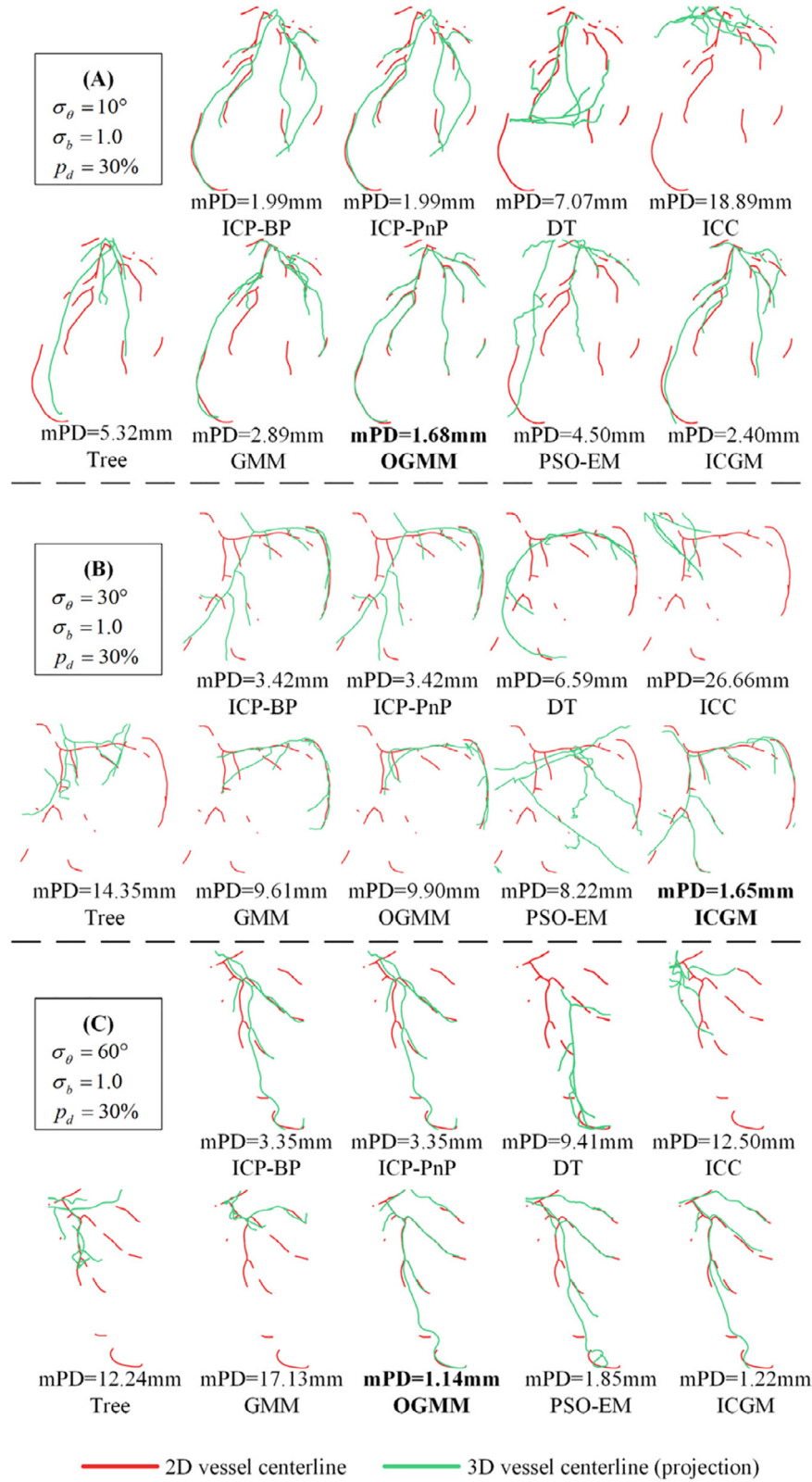


Fig. 5. Examples of non-rigid registration results of simulated data with combined defects (*Combine1*). 8 kinds of comparing methods and 3 kinds of ICGMs are tested. The parameters of simulated data are: (A) $\sigma_\theta = 10^\circ$, $\sigma_b = 1.0$, $p_d = 30\%$; (B) $\sigma_\theta = 30^\circ$, $\sigma_b = 1.0$, $p_d = 30\%$; (C) $\sigma_\theta = 60^\circ$, $\sigma_b = 1.0$, $p_d = 30\%$. Projected 3D and 2D centerlines are shown in green and red, respectively. The method and mean projected distance (short for mPD, unit: mm) are presented below the respective plotted sub-figures. The texts of registration with best accuracy are bolded.

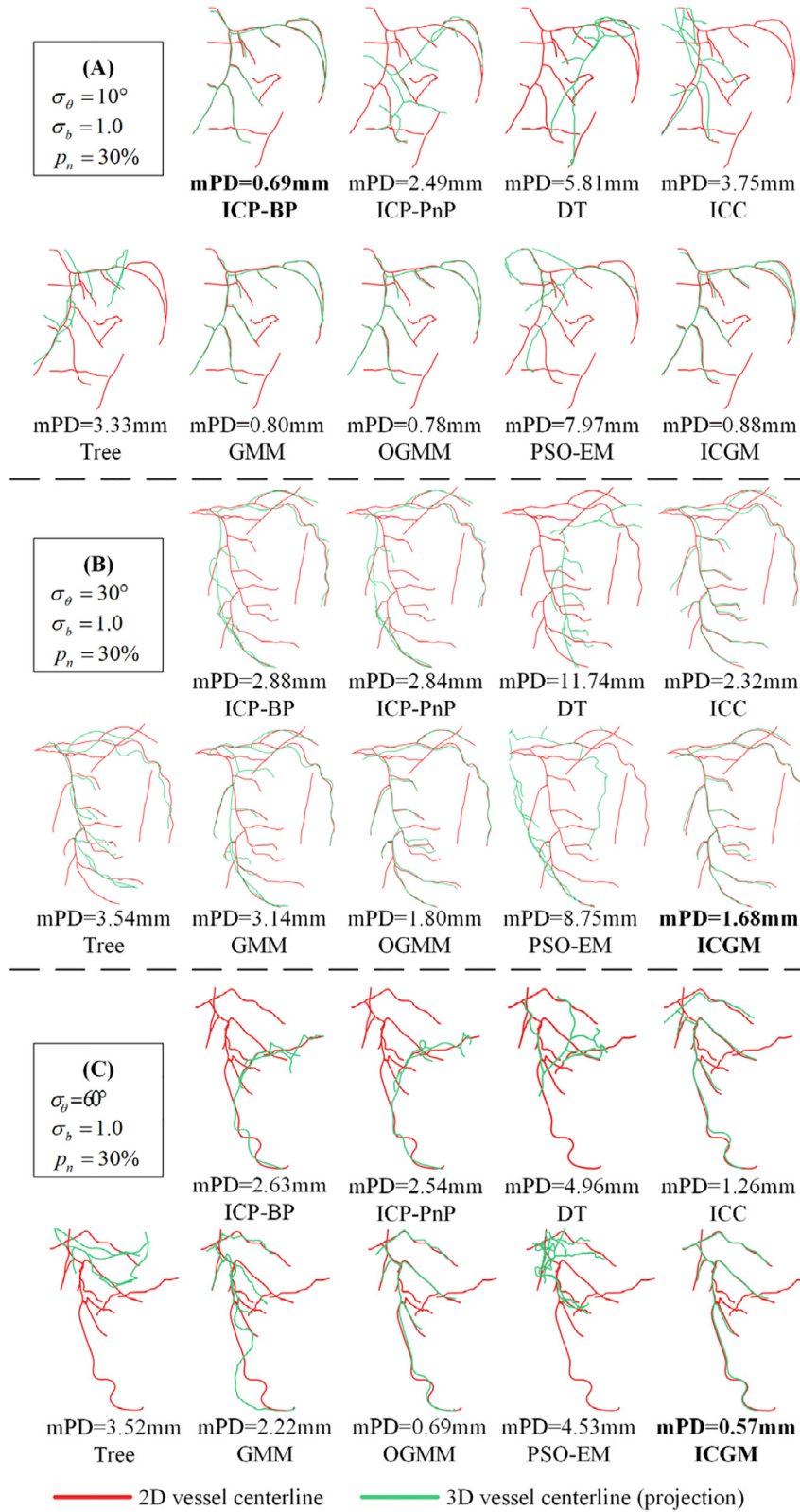


Fig. 6. Examples of non-rigid registration results of simulated data with combined defects (*Combine2*). 8 kinds of comparing methods and 3 kinds of ICGMs are tested. The parameters of simulated data are: (A) $\sigma_\theta = 10^\circ$, $\sigma_b = 1.0$, $p_n = 30\%$; (B) $\sigma_\theta = 30^\circ$, $\sigma_b = 1.0$, $p_n = 30\%$; (C) $\sigma_\theta = 60^\circ$, $\sigma_b = 1.0$, $p_n = 30\%$. Projected 3D and 2D centerlines are shown in green and red, respectively. The method and mean projected distance (short for mPD, unit: mm) are presented below the respective plotted sub-figures. The texts of registration with best accuracy are bolded.

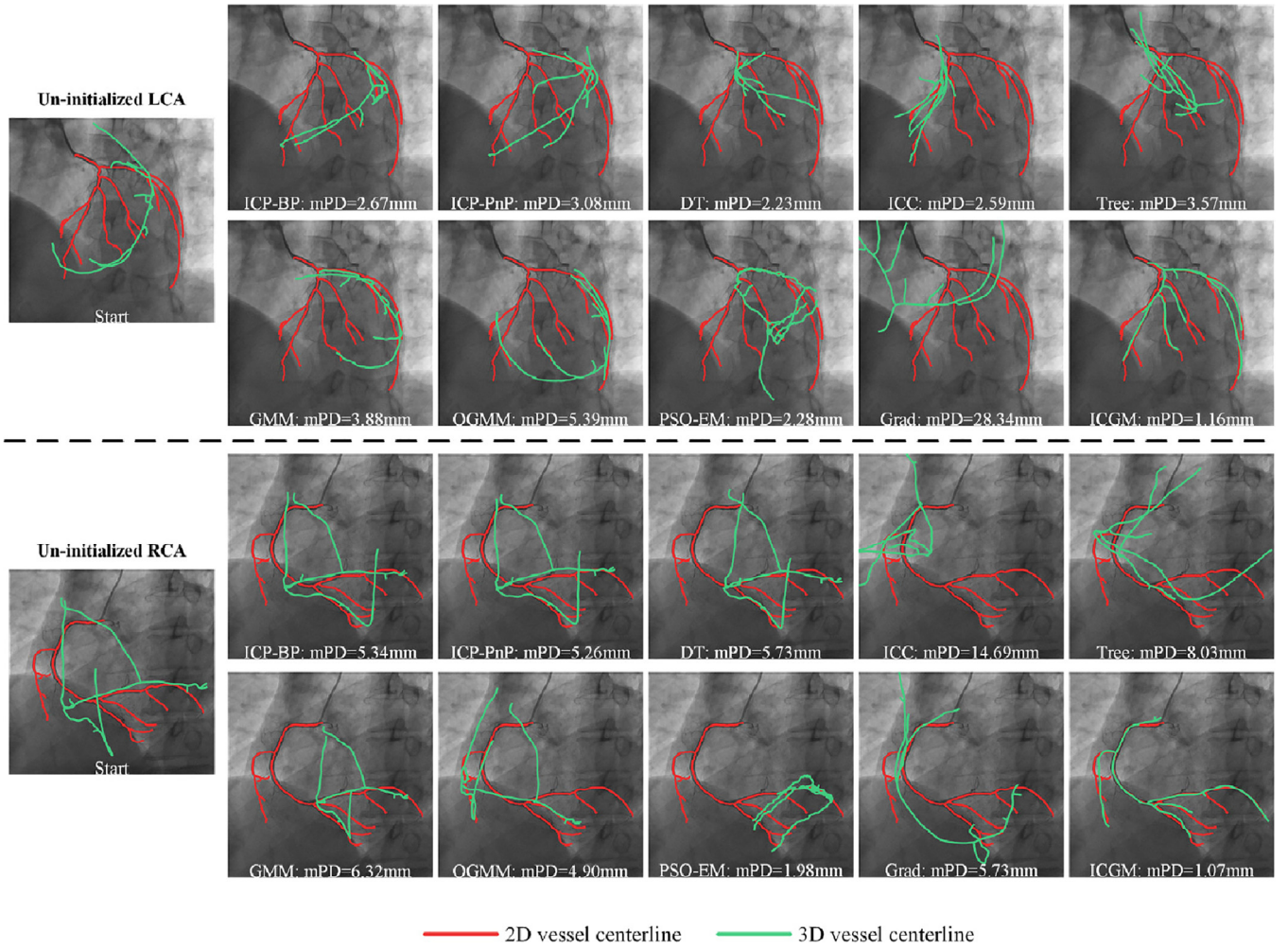


Fig. 7. Two examples of non-rigid 3D/2D registration results on un-initialized real data, including RCA (upon the dash line) and LCA (below the dash line). 9 kinds of comparing methods and 3 kinds of ICGMs are tested. Projected 3D centerlines are shown in green, while 2D centerlines are shown in red. The method and mean projected distance (short for mPD, unit: mm) are presented below the respective plotted sub-figures. The texts of registration with best accuracy are bolded and colored yellow.

tion, these methods predictably result in failed registration in most instances.

Therefore, we manually adjust the initial pose for each 2D vessel to perform 3D/2D registration on un-initialized and initialized data. The mean value and standard deviation of distance errors (mPD) of un-initialized data are 15.17mm and 7.35mm, respectively. After initialization, the mean value and standard deviation are 5.70mm and 3.14mm, respectively. The proposed method intends to find features crossing dimensions and modalities, such as topological structure, to achieve reliable correspondence and successful registration. The proposed method is expected to perform well on un-initialized and initialized data. A total of 30 pairs of 3D and 2D vessels are utilized in the real data experiments. Each pair of data is performed 4 times; that is, 240 registrations are performed for each method.

3. Results

The results are visualized as cumulative histograms of errors up to 3mm and examples of plotted vessel centerlines. The Y-axis value denotes the cumulative percentage of cases whose registration result under a certain value on X-axis. In simulation experiments, generated data with varying isolated defect (rotation angle differences, segment defects, branch noises and deformation) are

performed. For each set of parameters, 80 pairs of 3D and 2D vessel are generated. The results of simulated rigid and non-rigid registration experiments are shown in Fig. 3.

For rotation testing, the cumulative histogram curves decrease with increasing σ_θ , except those for PSO-EM and the proposed ICGM algorithm. In particular, ICGM exhibits excellent performance on data with large rotation differences. ICGM achieves a mean accuracy of 0.57mm in all the rotation tests. The other methods heavily rely on the initial pose, and their accuracies decrease distinctly with increasing σ_θ . PSO-EM uses global optimization to search for the optimal solution and can register data with large-scale rotation. However, its performance relies on the number of particles used. Moreover, the randomness of PSO-EM produces unstable results.

For segment defect testing, most methods are sensitive to missing points. As the defect p_d increases, the accuracy of ICGM decreases distinctly. Mean accuracies of 0.57mm, 1.58mm, and 2.22mm are achieved by the proposed method ICGM in the presence of 10%, 30%, and 50% defects, respectively. ICGM utilizes the topology of the vessel to achieve matching and registration. However, defects in the centerline seriously affect topology, which cause topology-based methods perform poorly. The ICC and Tree methods are sensitive to segment defects and achieve low accuracies. OGMM and GMM perform robustly against segment de-

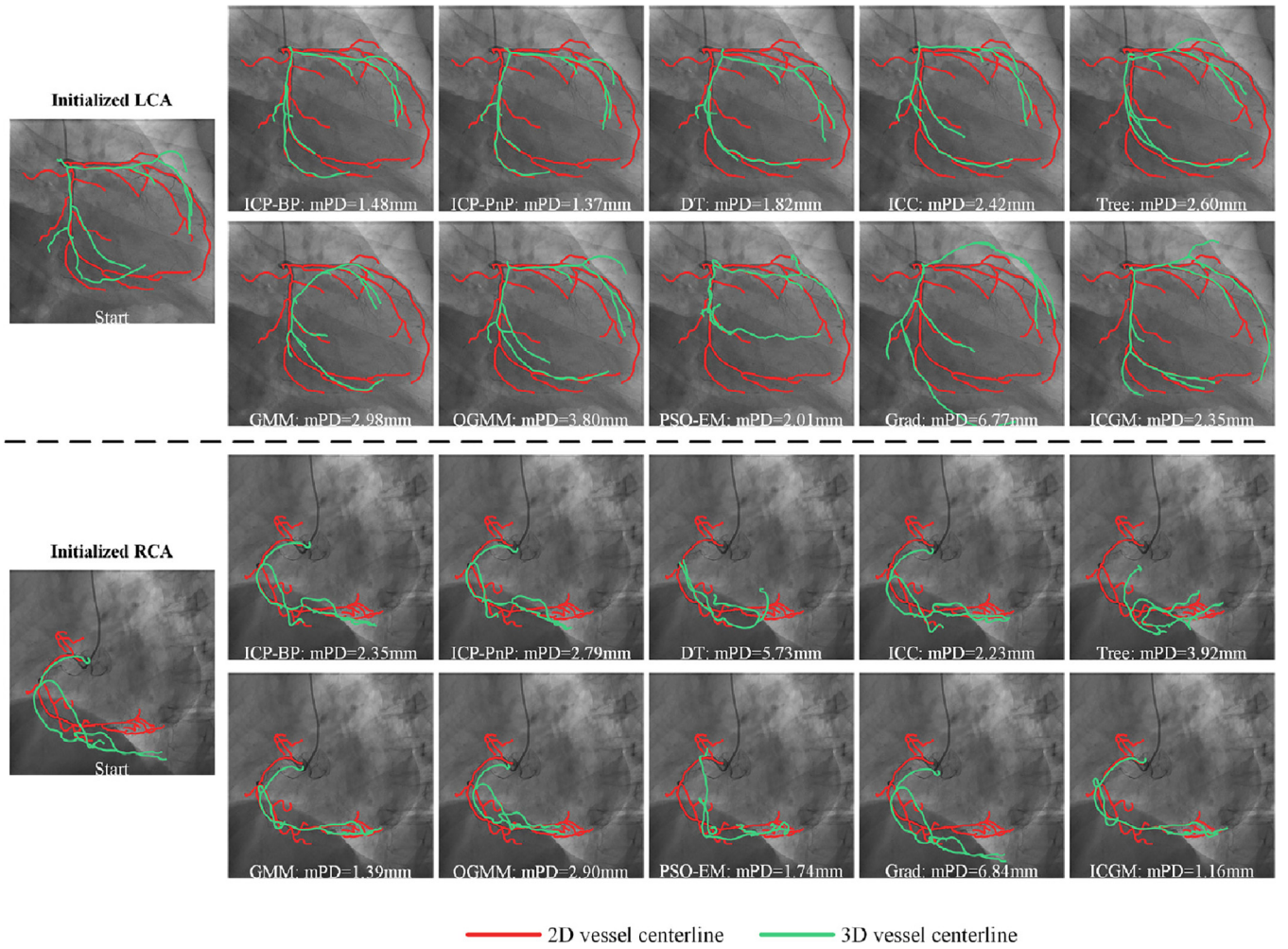


Fig. 8. Two examples of non-rigid 3D/2D registration results on initialized real data, including RCA (upon the dash line) and LCA (below the dash line). 9 kinds of comparing methods and 3 kinds of ICGMs are tested. Projected 3D centerlines are shown in green, while 2D centerlines are shown in red. The method and mean projected distance (short for mPD, unit: mm) are presented below the respective plotted sub-figures. The texts of registration with best accuracy are bolded and colored yellow.

fects. GMM and OGMM achieve mean accuracies of 1.30mm and 0.40mm, respectively, in the presence of 50% defects. The orientation used in OGMM is calculated from the perfect graph rather than from the defective graph.

For branch noise testing, most methods, except for the topology based method (ICC, Tree) and DT, are robust to branch noise. The proposed ICGM method is robust to branch noise and achieve mean accuracies of 0.85mm, 0.82mm and 0.74mm, respectively, even when 50% noise is added.

For deformation testing, we can conclude that all the methods, except for PSO and DT, perform steadily on deformed data. A similar deformation model is adopted for each registration method; hence, differences in their non-rigid registrations involve the matching strategies of matching-based methods (ICGM, ICP-PnP, ICP-BP, ICC, and Tree) or the cost functions of optimization-based methods (DT, GMM, OGMM, and PSO-EM). The proposed ICGM achieves 1.06mm mean accuracy at a deformation scale of 1.0.

Furthermore, simulation experiments on generated data with combined defects are performed. Two kinds of combinations of simulation parameters are adopted: 1) *Combine1*: 30% missed centerline points, $\sigma_b = 1.0$ for non-rigid deformation, $\{\sigma_\theta\} = \{10^\circ, 30^\circ, 60^\circ\}$ for rotation. 2) *Combine2*: 30% added branch noise, $\sigma_b = 1.0$ for non-rigid deformation, $\{\sigma_\theta\} = \{10^\circ, 30^\circ, 60^\circ\}$ for ro-

tation.. The results of simulated non-rigid registration experiments with combined defects are shown in Fig. 4. For *Combine1*, ICGM is less sensitive to varying rotation, while it performs with low accuracy for RGM relies on the completeness of vessel topology. OGMM performs best and is robust for missed centerline points, achieving a mean accuracy of 2.59mm for all registrations. For *Combine2*, ICGM is robust to branch noise even with rotation and deformation, achieving mean accuracies of 1.02mm, 1.09mm and 1.06mm for $\sigma_\theta = 10^\circ, 30^\circ$, and 60° respectively. Examples of non-rigid registrations simulated data generated with combined defects are presented in Figs. 5 and 6.

The results of experiments using real data are visualized by a plotted centerline in the XRA images and a cumulative histogram. In Fig. 7, two cases of registration of un-initialized real data are presented. The applied method and distance error are shown at the bottom of the respective plotted sub-figures. Large-scale rotation may exist in the initial pose of the un-initialized data, and only ICGM achieves good registration results in the two cases of real data. The results of two cases of initialized real data are shown in Fig. 8. As the initial pose is provided, most of methods are capable of registering the vessels with high accuracy. The cumulative histograms of errors on un-initialized and initialized data are shown in Fig. 9. The proposed ICGM is barely sensitive to the initial pose. Details on the registration errors of un-initialized and initial-

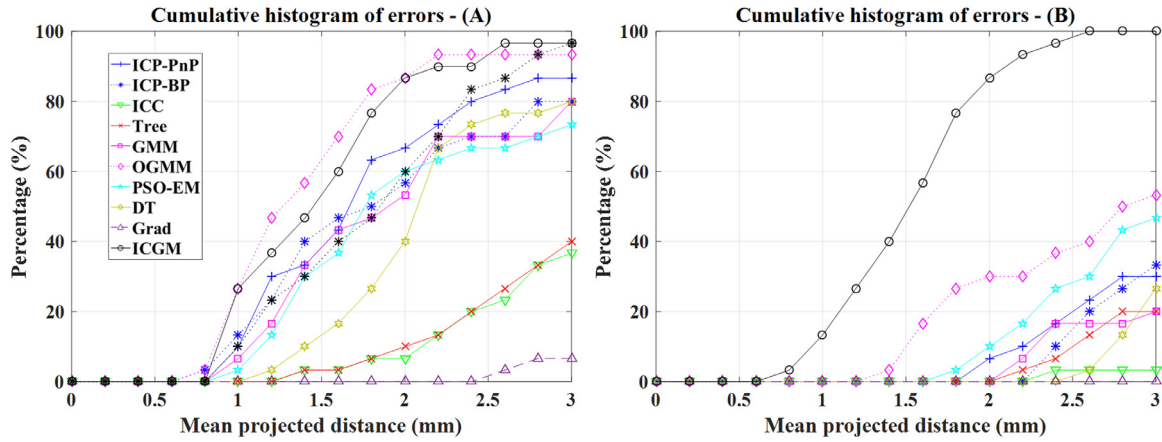


Fig. 9. Results of non-rigid registration experiments on (A) initialized and (B) un-initialized real data. 9 kinds of comparing methods and 3 kinds of ICGMs are tested. Percentage is indicated by Y-axis and 100% percentage indicates 120 times registrations. Mean projected distance (unit: mm) is indicated by X-axis.

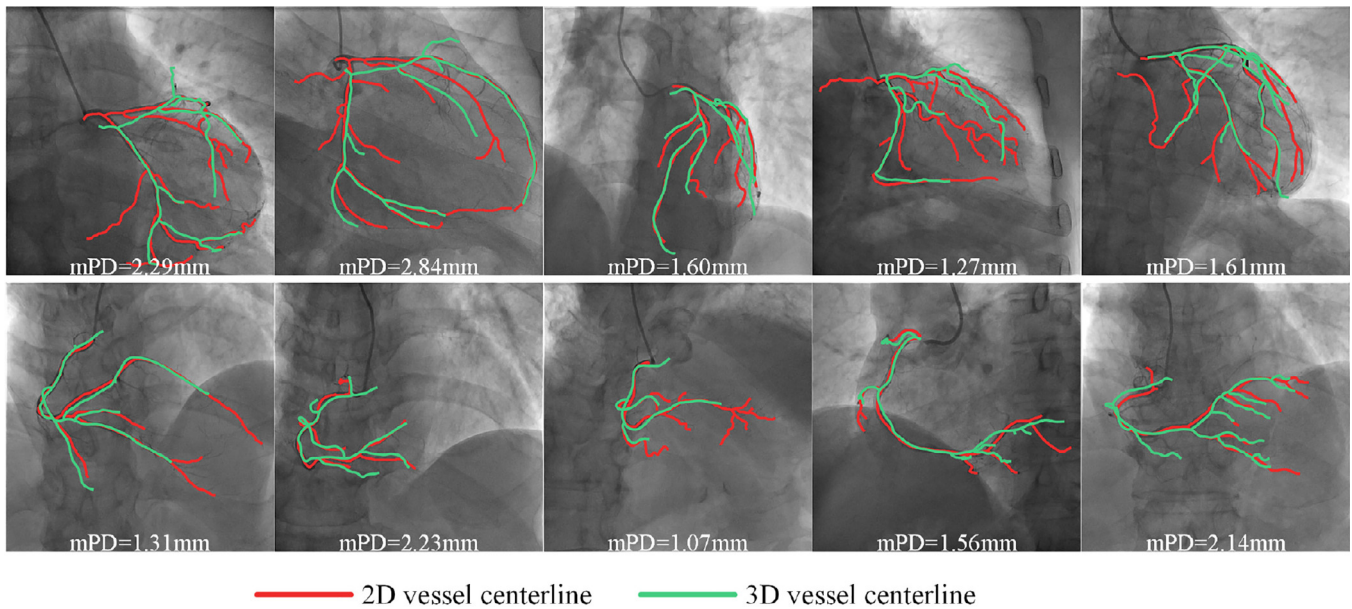


Fig. 10. Results Exhibition of ICGM. The first row indicates results of LCA and the second row indicates results of RCA.

Table IV

Registration errors of un-initialized and initialized real data (unit: mm), and elapsed time (unit: s).

	Mean Projected Distance (mm) and standard deviation				Average elapsed time (s)
	Un-initialized		Initialized		
	LCA	RCA	LCA	RCA	
ICP-PnP	3.53 (1.34)	4.40 (2.17)	1.36 (0.39)	2.49 (0.94)	21.43
ICP-BP	3.61 (1.45)	5.19 (2.98)	1.32 (0.39)	2.96 (1.23)	23.39
DT	7.97 (5.02)	4.31 (1.58)	2.02 (0.71)	3.26 (2.07)	2.21
ICC	6.27 (3.46)	11.16 (8.44)	2.89 (1.11)	6.05 (3.72)	94.96
Tree	6.64 (5.94)	8.47 (5.02)	2.87 (0.98)	4.54 (2.36)	48.80
GMM	16.91 (13.60)	11.31 (10.14)	1.81 (1.61)	4.82 (7.42)	23.54
OGMM	6.54 (9.37)	5.16 (7.31)	1.15 (0.29)	1.78 (0.80)	35.86
PSO-EM	6.40 (7.93)	8.70 (10.32)	2.12 (1.66)	4.74 (7.50)	381.34
Grad	21.82 (7.43)	17.15 (8.16)	10.27 (9.08)	11.75 (8.54)	13.95
ICGM	1.59 (0.40)	1.57 (0.45)	1.55 (0.54)	1.59 (0.73)	21.35

ized real data are presented in Table IV. The values in brackets are the standard deviation of distance errors. The codes are running on a PC (Intel I7-7700 CPU, 32GB RAM, MATLAB platform), and the elapsed times of all method are presented in Table IV. At last, 10 cases of ICGM results are exhibited in Fig. 10, including 5 cases of LCA and 5 cases of RCA.

4. Conclusion and discussion

An ICGM framework for 3D/2D vessel registration is proposed. Experiments on simulated and real datasets are performed compared with nine other methods, including ICP-PnP, ICP-BP [7], DT [8], ICC [18], Tree [22], 3D/2D GMM [20], OGMM [7], PSO-EM

[9] and Grad [11]. The simulation experiments are designed to test the robustness with respect to the initial pose, missed vessel points, branch noise, and deformation. The registration results of the simulated data with isolated and combined defects are presented in Figs. 3 and 4, respectively. Expectedly, the proposed method is barely sensitive to the initial pose even with other combined defects.

The proposed RGM can match two graphs with large-scale transformation. Given the correspondences of 3D and 2D points, a closed-form solution for PnP is used to obtain accurate rigid transformation. In addition, the iterative matching framework is helpful in progressively achieving accurate registration. Experiments on real data demonstrate a similar conclusion, i.e., the proposed method is un-sensitive to initial pose. In the coarse matching of vessel graph nodes, some node correspondences may be inaccurate. For the restriction of branch direction in the vessel graph, ICGM will not produce inaccurate correspondences of vessel points. The detailed registration errors and computation times of all the methods are presented in Table IV.

Among the 3D/2D registration methods, ICGM and PSO-EM can align data with large-scale rigid transformation. PSO-EM utilizes global optimization to obtain high accuracy; nonetheless, it is quite time-consuming, requiring an average of 381.34s for real data. OGMM achieves the highest accuracy on simulated data with low rotation difference and initialized real data due to the vessel orientation used to estimate L_2 distance. ICP-PnP and ICP-BP also perform well on data with initialization. However, they are sensitive to branch noise and missing points. DT precomputes a distance transform of the 2D centerline. Furthermore, it performs the fastest among the methods, requiring an average of 2.21s for real data. Transformation is calculated via optimization instead of closed-form solution used in ICP-PnP and ICP-BP. Thus, DT is more sensitive to noise and defect than ICP-PnP and ICP-BP. DT was originally proposed for registering 3D vessel to bi-plane 2D vessels. Meanwhile, our experiments are performed on mono-plane XRA. The ICC and Tree methods utilize the topology continuity of vessels to obtain accurate matching. Consequently, they are sensitive to missing vessel points. ICGM is also sensitive to missing vessel points; nonetheless, it is more robust than ICC and Tree methods due to the global matching strategy of RGM. Grad is proposed to rigidly register cerebral angiograms. It performs worst on coronary angiograms. However, the comparison works are biased to a certain extent. Since the nine competing methods are re-implemented according to their paper or provided by the authors. The methods are not tuned for large capture range. These methods could eventually be fine-tuned to include an initial pose estimation procedure.

The robustness to noise and deformation and accuracy are the primary advantages of ICGM. Moreover, ICGM is insensitive to the initial pose of registration. By contrast, the dependence on completed vessel topology is a limitation of ICGM. The vessel centerlines are the input of ICGM, therefore the errors of centerline extraction and vessel segmentation will influence the final registration results. Besides, ICGM is sensitive to topological defect. As expected, an over-segmentation of XRA performs better than an under-segmentation for the proposed method. Moreover, several comparative methods are based on the re-implementation on our datasets, thereby resulting in personal bias. However, the comparison experiments can still validate the effectiveness of the proposed method to a certain extent.

Declaration of Competing Interest

The authors have declared that no conflict of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.cmpb.2020.105901.

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